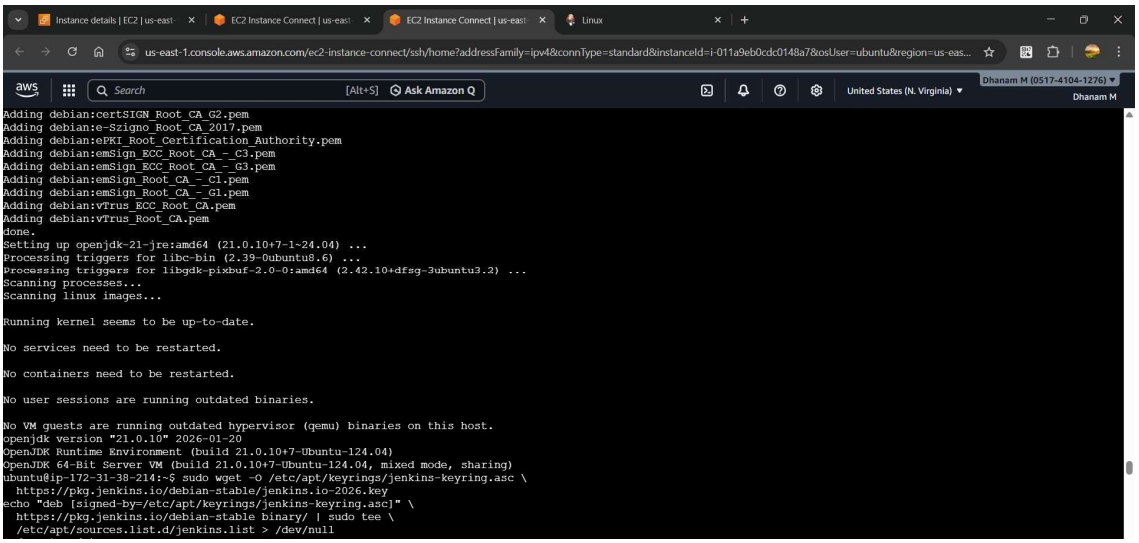
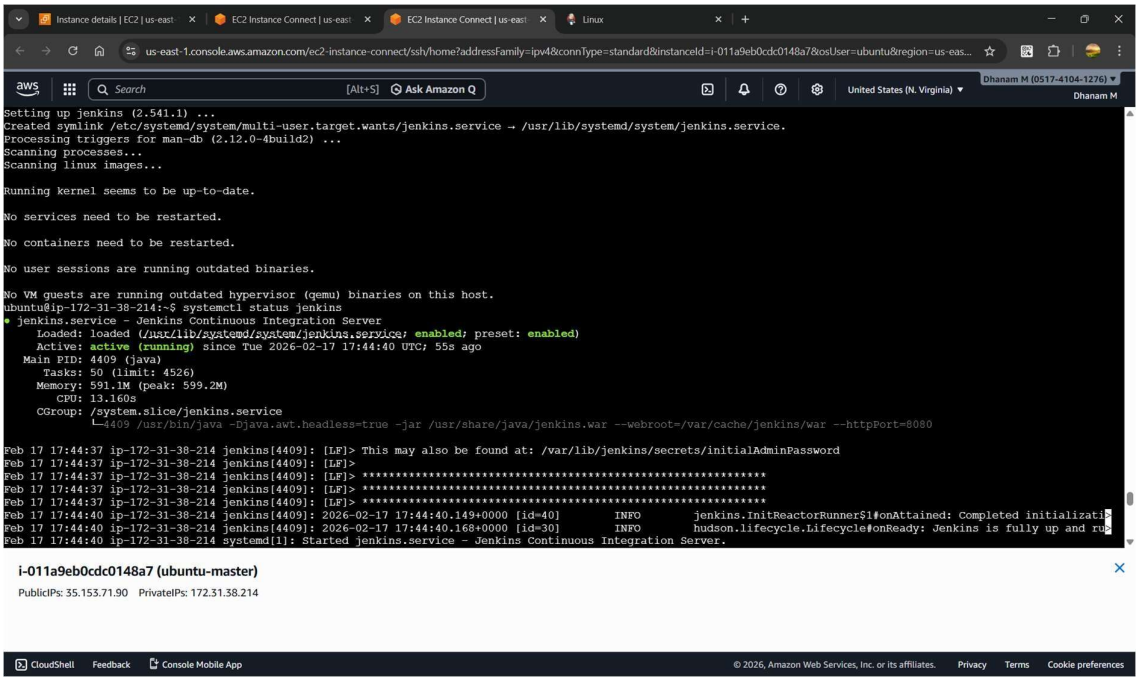
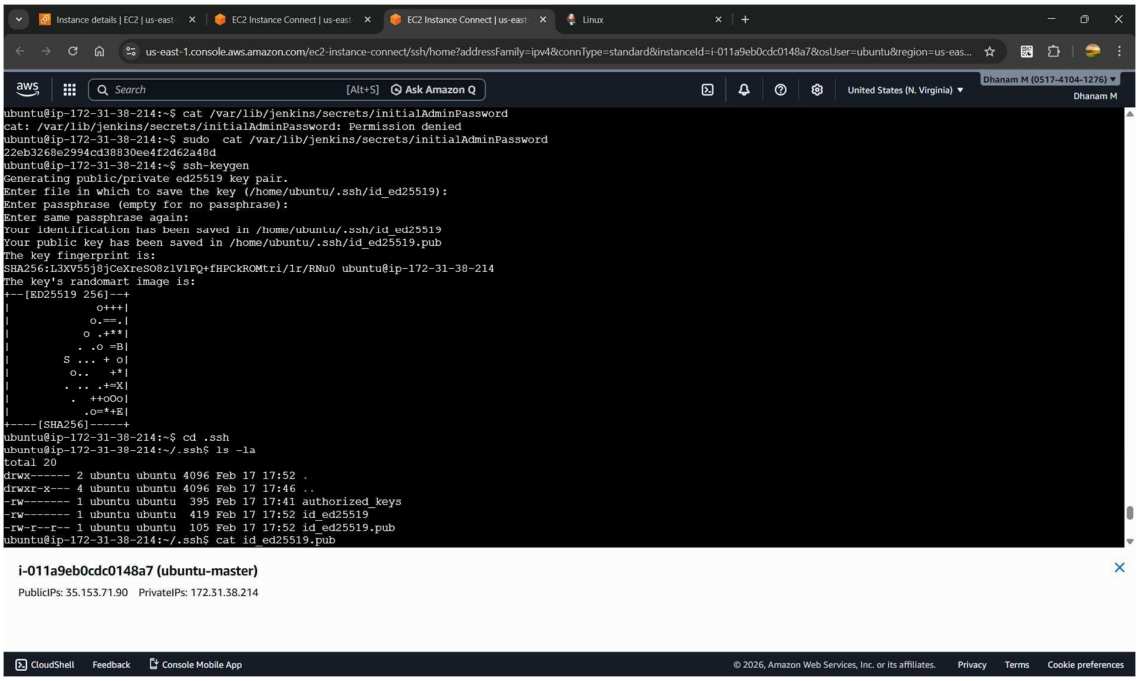
Jenkins master slave

- Install the java and Jenkins in ubuntu master. Install java only in ubuntu slave.

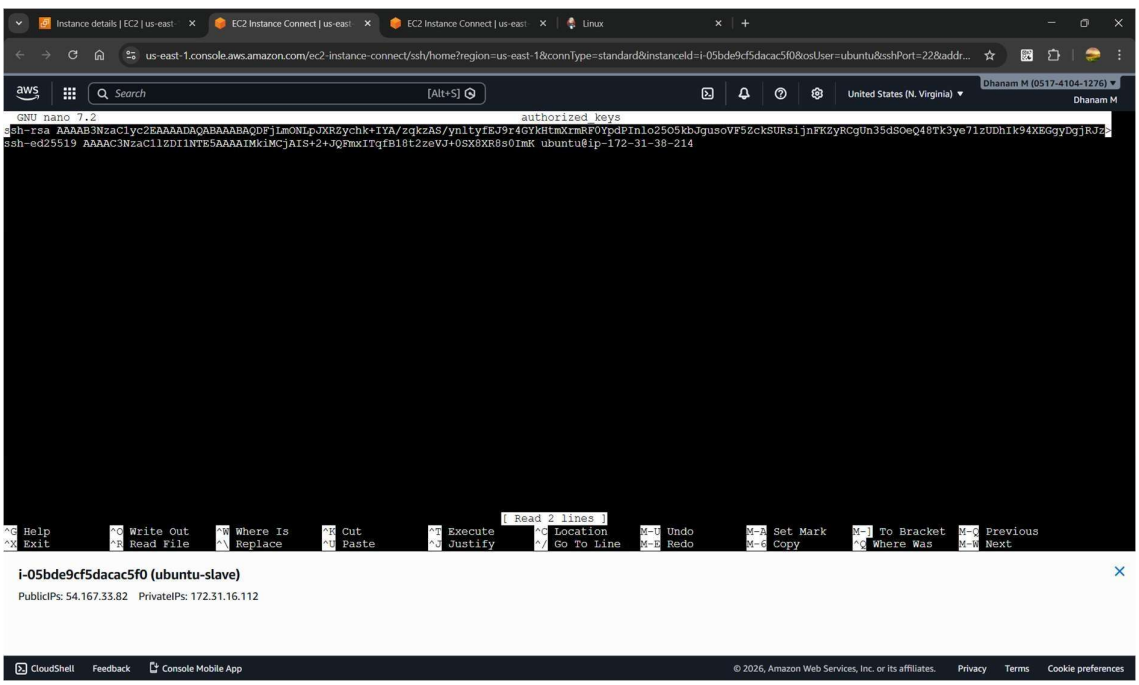




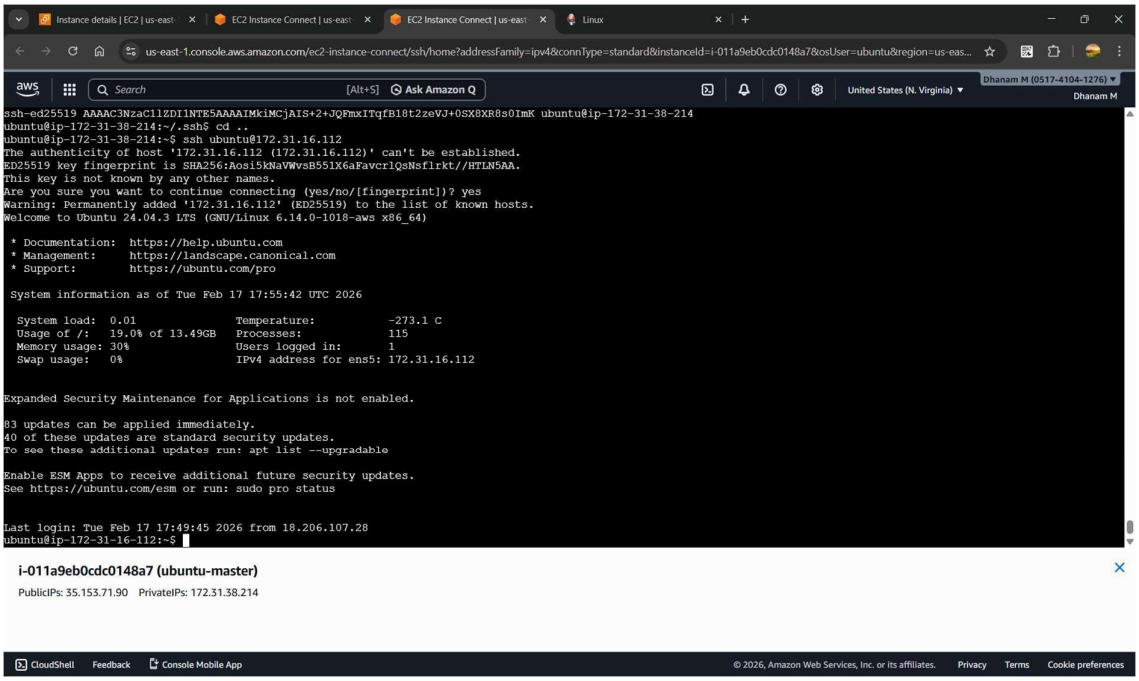
- Using “ssh-keygen” to generate public key and private key in “.ssh” directory.



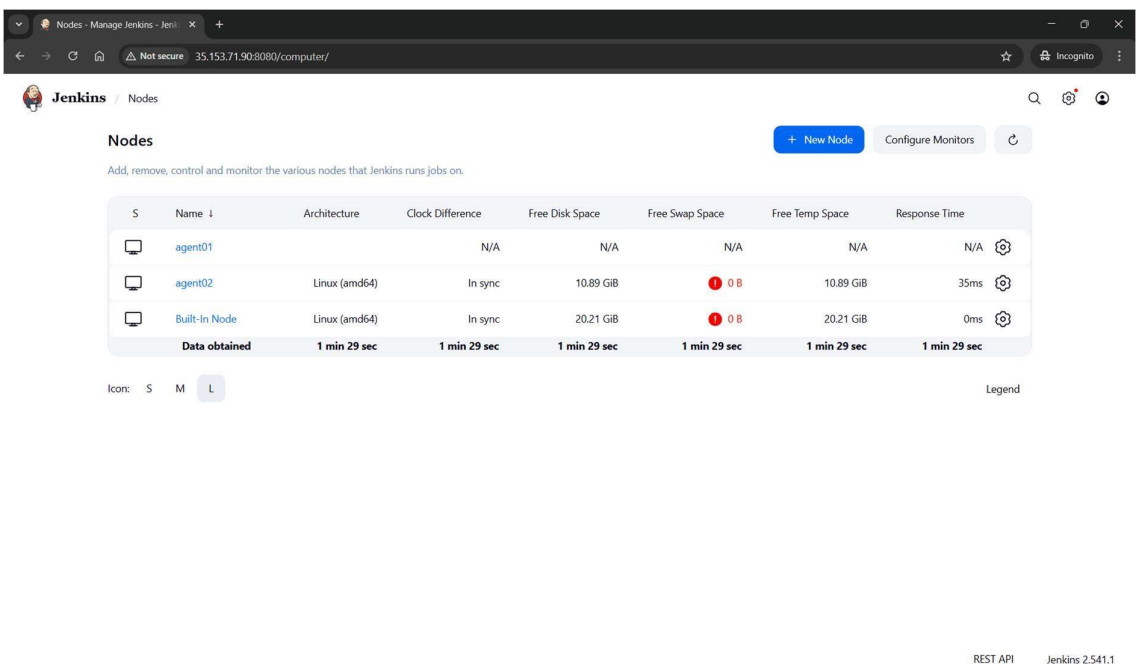
- using “id_ED22559.pub” to “cat” of an inside the public key to copy the key go to slave “nano authorized key” to paste the public key.



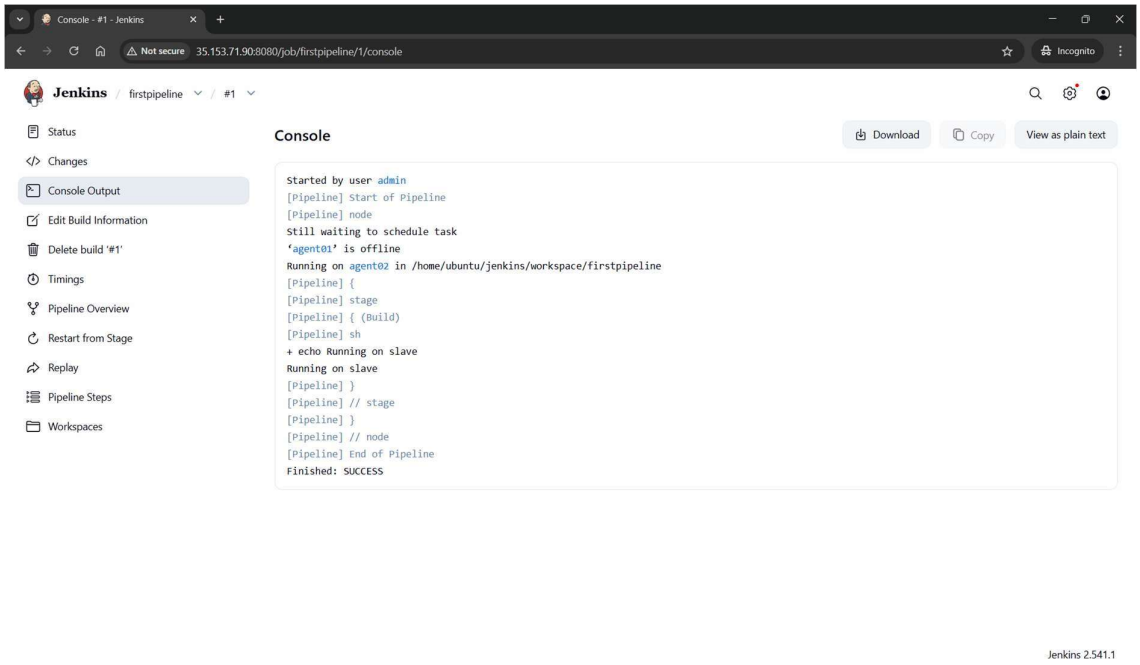
- the ubuntu master connected to slave to monitor and schedule the job.

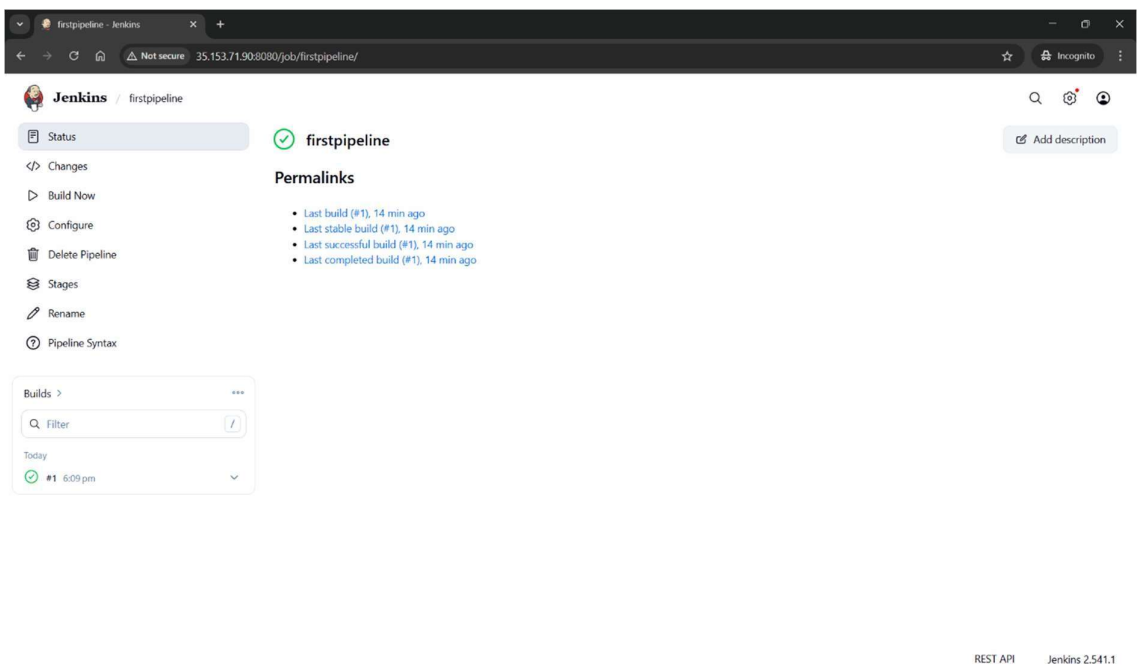


- create the node in Jenkins build in status.

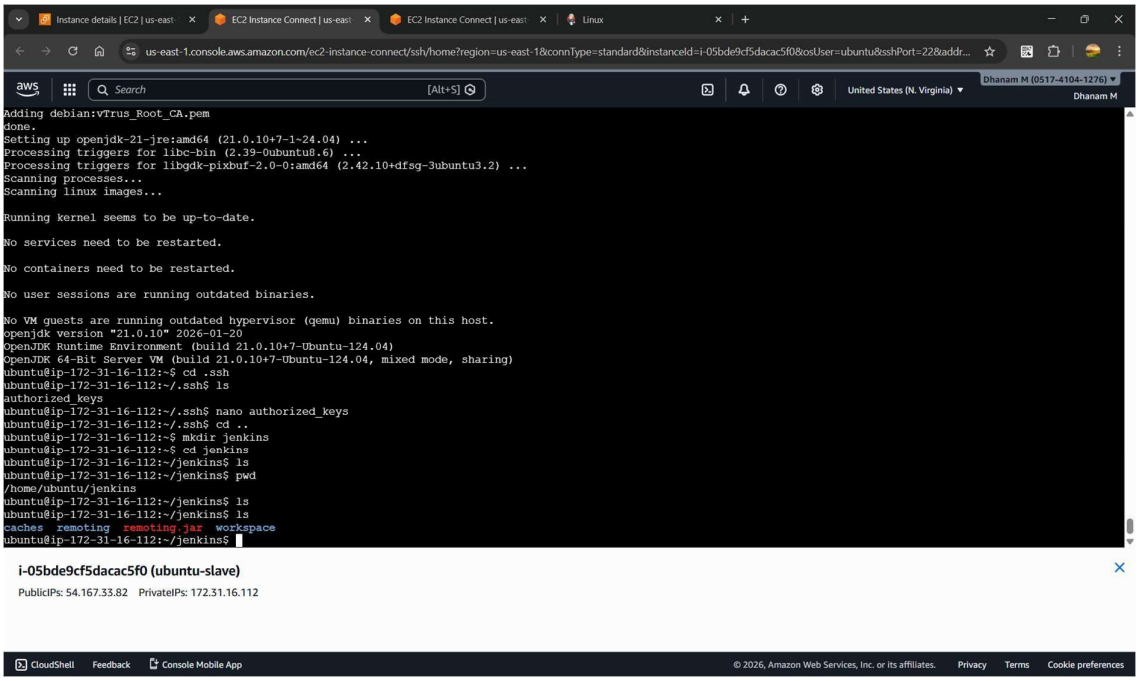


- to build in the pipeline.





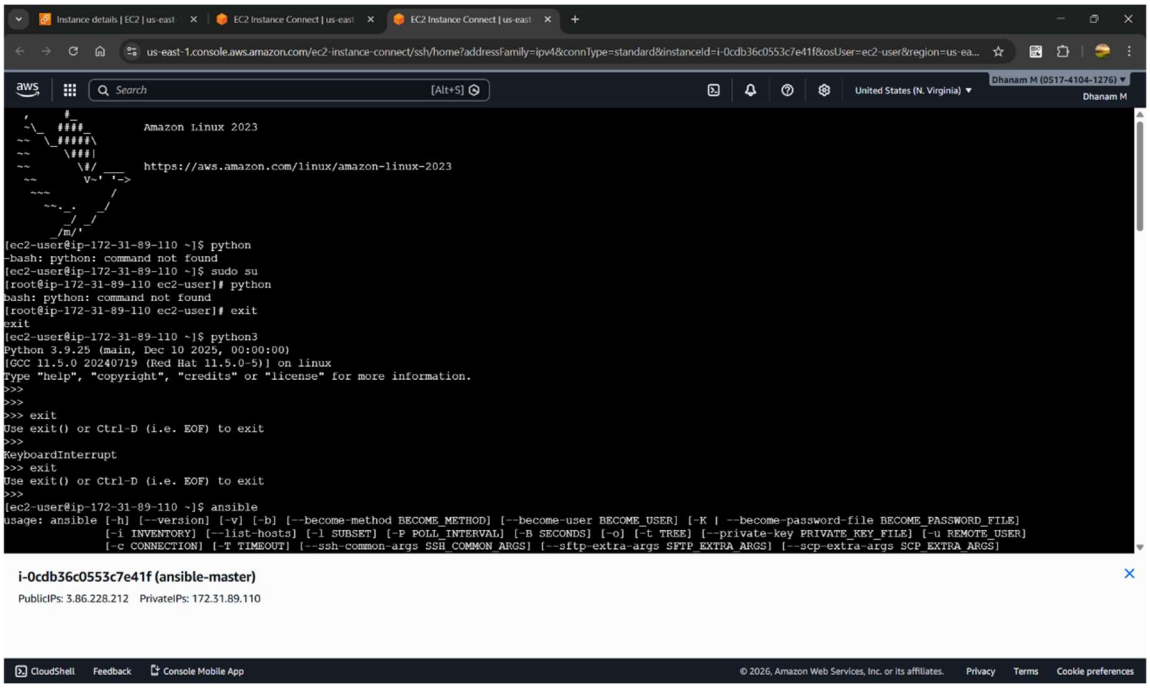
- run the pipeline successfully.



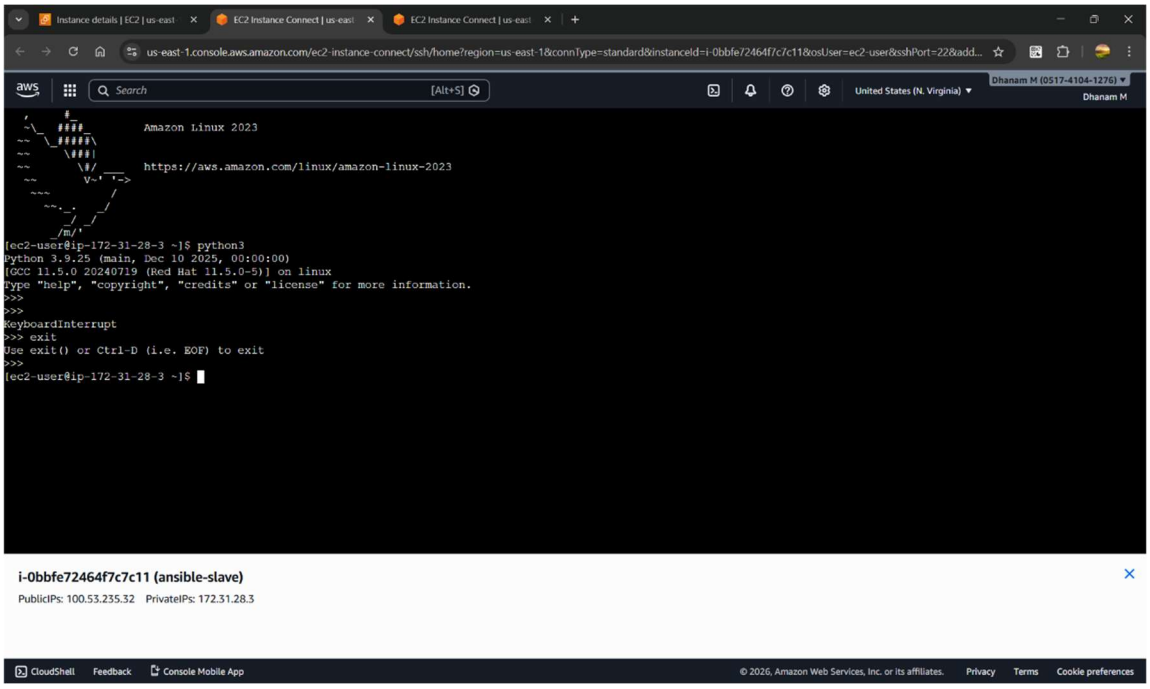
- creating remoting jar and workspace in Jenkins directory.

Ansible Master Slave

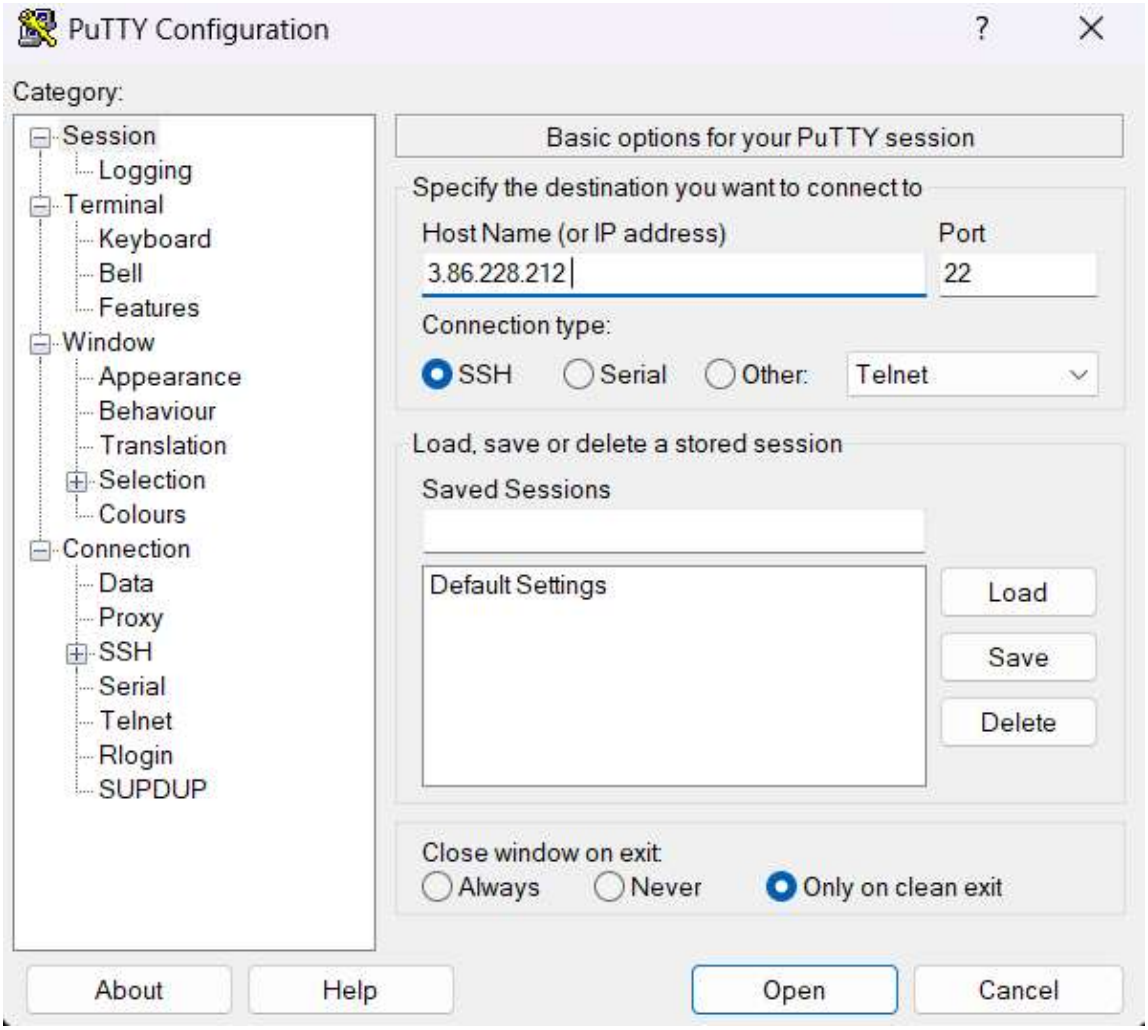
- ansible master to install an python3 -pip and install an ansible using an amazon linux machine.



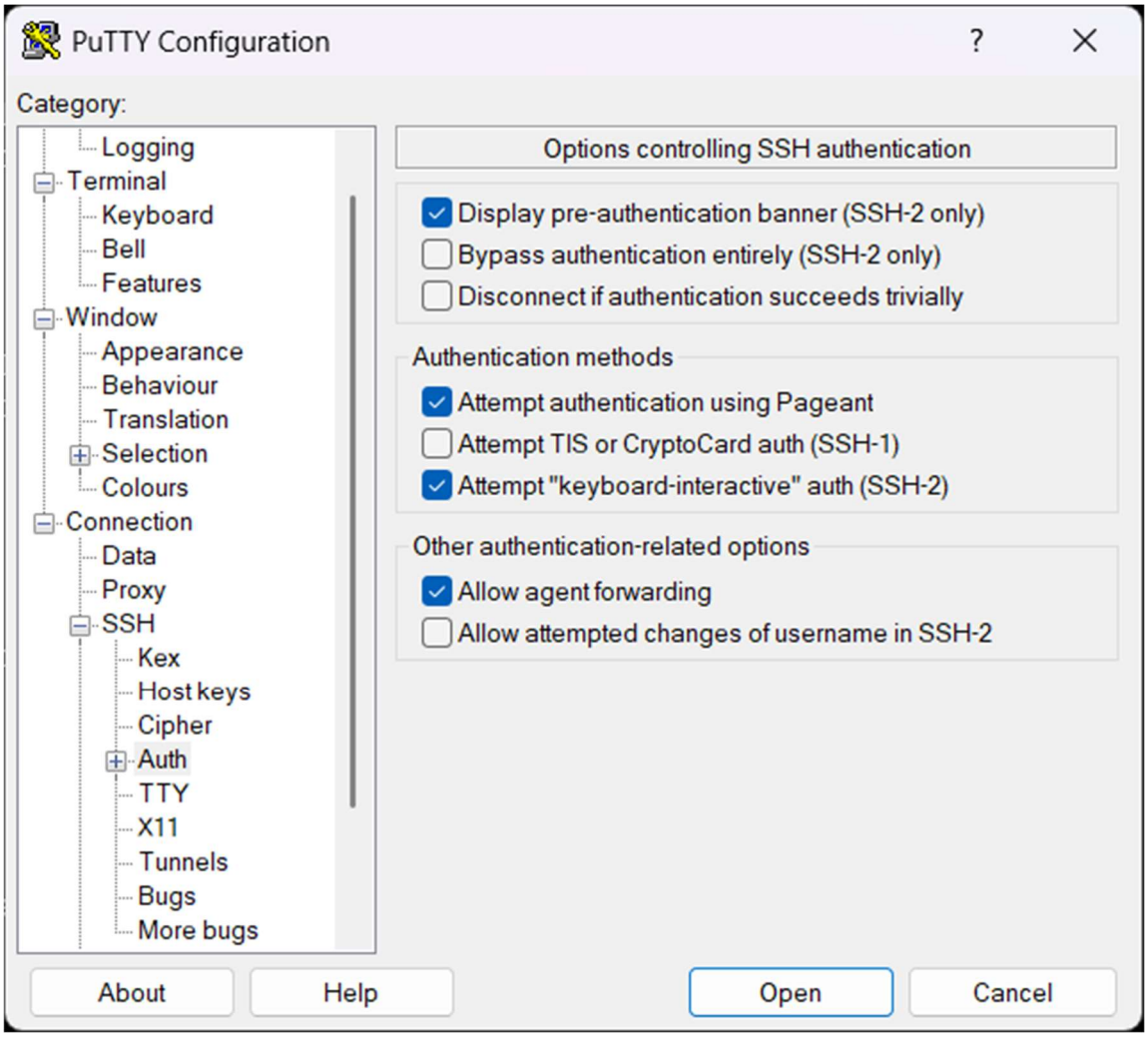
- ansible slave to install python3-pip using an amazon linux.



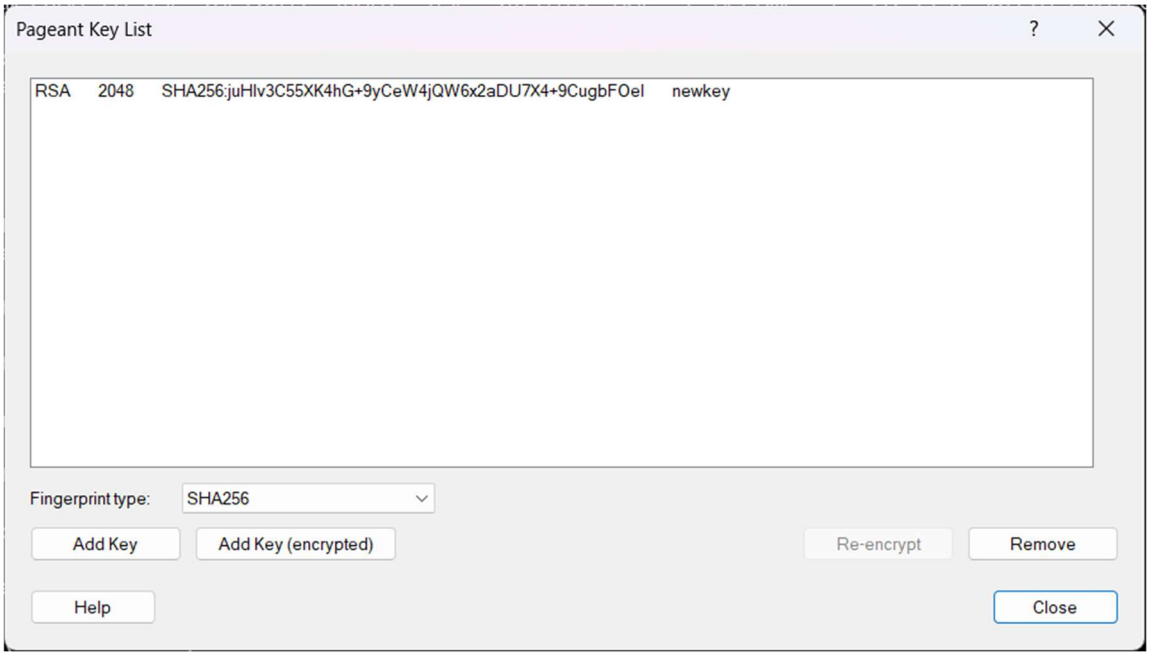
- to download an putty.exe and open an putty to copy master public ip to paste host in putty.



- to click an ssh and open to click authorized and allow agent forwarding and click open.



- to open pagent.ex to add key ppk.key and close.



- login ec2-user.

- install httpd using ansible machine.

```
[ec2-user@ip-172-31-89-110 ~]$ ansible --version
ansible [core 2.15.3]
  config file = /home/ec2-user/ansible.cfg
  configured module search path = ['/home/ec2-user/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  ansible collection location = /home/ec2-user/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.9.25 (main, Dec 10 2025, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3.9)
  jinja version = 3.1.4
  libyaml = True
[ec2-user@ip-172-31-89-110 ~]$ ansible localhost -m yum -a "name=httpd state=present" -b -i localhost,
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning
of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
localhost | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "msg": "",
    "rc": 0,
    "results": [
        "Installed: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch",
        "Installed: httpd-core-2.4.66-1.amzn2023.0.1.x86_64",
        "Installed: mailcap-2.1.49-3.amzn2023.0.3.noarch",
        "Installed: apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64",
```

ec2-user@ip-172-31-89-110:~

```
    "changed": false,
    "ping": "pong"
}
[ec2-user@ip-172-31-89-110 ~]$ ansible --version
ansible [core 2.15.3]
  config file = /home/ec2-user/ansible.cfg
  configured module search path = ['/home/ec2-user/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3.9/site-packages/ansible
  ansible collection location = /home/ec2-user/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.9.25 (main, Dec 10 2025, 00:00:00) [GCC 11.5.0 20240719 (Red Hat 11.5.0-5)] (/usr/bin/python3.9)
  jinja version = 3.1.4
  libyaml = True
[ec2-user@ip-172-31-89-110 ~]$ ansible localhost -m yum -a "name=httpd state=present" -b -i localhost,
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning
of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
localhost | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "msg": "",
    "rc": 0,
    "results": [
        "Installed: httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch",
        "Installed: httpd-core-2.4.66-1.amzn2023.0.1.x86_64",
        "Installed: mailcap-2.1.49-3.amzn2023.0.3.noarch",
        "Installed: apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64",
        "Installed: mod_lua-2.4.66-1.amzn2023.0.1.x86_64",
        "Installed: apr-1.7.5-1.amzn2023.0.4.x86_64",
        "Installed: httpd-tools-2.4.66-1.amzn2023.0.1.x86_64",
        "Installed: apr-util-1.6.3-1.amzn2023.0.2.x86_64",
        "Installed: mod_http2-2.0.27-1.amzn2023.0.3.x86_64",
        "Installed: httpd-2.4.66-1.amzn2023.0.1.x86_64",
        "Installed: generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch",
        "Installed: apr-util-lmdb-1.6.3-1.amzn2023.0.2.x86_64",
        "Installed: libbrotli-1.0.9-4.amzn2023.0.2.x86_64"
    ]
}
[ec2-user@ip-172-31-89-110 ~]$ ansible localhost -m service -a "name=httpd state=started enabled=yes" -b -i localhost,
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.9, but future installation of another Python interpreter could change the meaning
of that path. See https://docs.ansible.com/ansible-core/2.15/reference_appendices/interpreter_discovery.html for more information.
localhost | CHANGED => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.9"
    },
    "changed": true,
    "enabled": true,
    "name": "httpd",
    "state": "started",
    "status": {
        "AccessSELinuxContext": "system_u:object_r:httpd_unit_file_t:s0",
        "ActiveEnterTimestampMonotonic": "0",
        "ActiveExitTimestampMonotonic": "0",
        "ActiveState": "inactive",
```

It works!

- run the host <public ip>:httpd port number 80.